

CS257 Linear and Convex Optimization

Homework 4

Due: October 19, 2020

October 12, 2020

1. Determine if the following functions are convex, concave, or neither.

(a). $f(\mathbf{x}) = f(x_1, x_2, x_3) = 2x_1^2 + x_1x_3 + x_2^2 + 2x_2x_3 + \frac{1}{2}x_3^2$ on $\mathbb{R}^3$

(b). $f(\mathbf{x}) = f(x_1, x_2) = (x_1x_2)^{-1}$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(c). $f(x_1, x_2) = x_1x_2$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(d). $f(x_1, x_2) = \frac{x_1}{x_2}$ on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

(e). $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, where $0 \leq \alpha \leq 1$, on $\mathbb{R}_{++}^2 = \{(x_1, x_2) : x_1 > 0, x_2 > 0\}$

Solution. (a). The Hessian matrix is

$$H = \nabla^2 f(\mathbf{x}) = \begin{pmatrix} 4 & 0 & 1 \\ 0 & 2 & 2 \\ 1 & 2 & 1 \end{pmatrix}$$

Check the leading principal minors.

$$D_1(H) = \det(4) > 0, \quad D_2(H) = \det \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix} = 8, \quad D_3(H) = \det H = 4 \det \begin{pmatrix} 2 & 2 \\ 2 & 1 \end{pmatrix} + \det \begin{pmatrix} 0 & 2 \\ 1 & 2 \end{pmatrix} = -10$$

so H is not positive semidefinite. It is neither negative semidefinite since $D_2(-H) = -4$. Therefore, f is neither convex nor concave.

(b). The Hessian matrix is

$$H = \nabla^2 f(\mathbf{x}) = \begin{pmatrix} \frac{2}{x_1^3 x_2} & \frac{1}{x_1^2 x_2^2} \\ \frac{1}{x_1^2 x_2^2} & \frac{2}{x_1 x_2^3} \end{pmatrix}$$

Check the leading principal minors.

$$D_1(H) = \frac{2}{x_1^3 x_2} > 0, \quad D_2(H) = \det H = \frac{3}{x_1^4 x_2^4} > 0$$

So H is positive definite and f is convex.

(c). The Hessian matrix is

$$H = \nabla^2 f(\mathbf{x}) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Check the principal minors.

$$\det H_{11} = \det -H_{11} = 0, \quad \det H_{22} = \det -H_{22} = 0, \quad \det H = \det -H = -1 < 0$$

So H is neither positive semidefinite nor negative semidefinite, and f is neither convex nor concave.

(d). The Hessian matrix is

$$H = \nabla^2 f(\mathbf{x}) = \begin{pmatrix} 0 & -\frac{1}{x_2^2} \\ -\frac{1}{x_2^2} & \frac{2x_1}{x_2^3} \end{pmatrix}$$

Check the principal minors.

$$\det H = \det(-H) = -\frac{1}{x_2^4} < 0$$

So H is neither positive semidefinite nor negative semidefinite and f is neither convex nor concave.

(e). The Hessian matrix is

$$H = \nabla^2 f(\mathbf{x}) = \begin{pmatrix} \alpha(\alpha-1)x_1^{\alpha-2}x_2^{1-\alpha} & \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} \\ \alpha(1-\alpha)x_1^{\alpha-1}x_2^{-\alpha} & \alpha(\alpha-1)x_1^\alpha x_2^{-\alpha-1} \end{pmatrix}$$

Check the principal minors.

$$\det -H_{11} \geq 0, \quad \det -H_{22} \geq 0, \quad \det -H = 0$$

So H is negative semidefinite and f is concave.

□

2. Prove the following statements.

(a). $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^4$ is strictly convex over $\mathbb{R}$.

(b). $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $g(x_1, x_2) = x_1^2 + x_2^4$ is strictly convex over $\mathbb{R}^2$.

Hint: Use the first-order condition.

Proof. (a). For any $x \in \mathbb{R}$ and $0 \neq d \in \mathbb{R}$,

$$f(x+d) - f(x) - f'(x)d = (x+d)^4 - x^4 - 4x^3d = 6d^2 \left[\left(x + \frac{1}{3}d\right)^2 + \frac{1}{18}d^2 \right] \geq \frac{1}{3}d^4 > 0$$

By the first-order condition, f is strictly convex.

(b). For any $\mathbf{x} \in \mathbb{R}^2$ and $\mathbf{0} \neq \mathbf{d} \in \mathbb{R}^2$

$$\begin{aligned} g(\mathbf{x} + \mathbf{d}) - g(\mathbf{x}) - \nabla g(\mathbf{x})^T \mathbf{d} &= (x_1 + d_1)^2 + (x_2 + d_2)^4 - x_1^2 - x_2^4 - 2x_1d_1 - 4x_2^3d_2 \\ &= d_1^2 + [f(x_2 + d_2) - f(x_2) - f'(x_2)d_2] \geq d_1^2 + \frac{1}{3}d_2^4 > 0 \end{aligned}$$

By the first-order condition, f is strictly convex.

□

3. Suppose $f : \mathbb{R}^n \rightarrow (-\infty, +\infty]$ satisfies Jensen's inequality

$$f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y}), \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^n, \theta \in [0, 1]$$

Show that the domain of f

$$\text{dom } f = \{\mathbf{x} \in \mathbb{R}^n : f(\mathbf{x}) < +\infty\}$$

is convex.

Proof. Let $\mathbf{x}, \mathbf{y} \in \text{dom } f$. Then $f(\mathbf{x}) < +\infty, f(\mathbf{y}) < +\infty$. Since $\theta, \bar{\theta} \in [0, 1]$,

$$-\infty \leq \theta f(\mathbf{x}) < +\infty, \quad -\infty \leq \bar{\theta} f(\mathbf{y}) < +\infty$$

By Jensen's inequality,

$$f(\theta \mathbf{x} + \bar{\theta} \mathbf{y}) \leq \theta f(\mathbf{x}) + \bar{\theta} f(\mathbf{y}) < +\infty$$

so $\theta \mathbf{x} + \bar{\theta} \mathbf{y} \in \text{dom } f$. Therefore, $\text{dom } f$ is convex. □

4. Is the following set convex? Show your argument.

$$S = \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x} > \mathbf{0}, x_1 \log x_1 + x_2 \log x_2 \leq 2\}$$

Solution. The negative entropy $h(x) = x \log x$ is convex over $(0, \infty)$. $g_i(\mathbf{x}) = \mathbf{e}_i^T \mathbf{x}$, $i = 1, 2$, are affine functions, where $\mathbf{e}_1 = (1, 0)^T, \mathbf{e}_2 = (0, 1)^T$. So $f_i(\mathbf{x}) = h(g_i(\mathbf{x})) = x_i \log x_i$ are convex over $\{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x} > \mathbf{0}\}$, and so is $f = f_1 + f_2$. The set S is convex since is the sublevel set of f , $S = \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x} > \mathbf{0}, f(\mathbf{x}) \leq 2\}$.

Alternatively, you can show f is convex by checking its Hessian

$$\nabla^2 f(\mathbf{x}) = \text{diag}\{x_1^{-1}, x_2^{-2}\} \succ \mathbf{O}$$

for $\mathbf{x} > \mathbf{0}$. □

5. Determine if the following optimization problems are convex optimization.

(a).

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2 \\ \text{s.t.} \quad & (x_1 - x_2)^2 + 4x_1x_2 + e^{x_1+x_2} \leq 0 \\ & x_1 - 3x_2 = 0 \end{aligned}$$

Proof. For $f_0(\mathbf{x}) = x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2$, its Hesse $\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$ is semidefinite, thus $f_0(\mathbf{x})$ is convex.

For $f_1(\mathbf{x}) = (x_1 - x_2)^2 + 4x_1x_2 + e^{x_1+x_2}$, its Hesse: $\begin{pmatrix} 2 + e^{x_1+x_2} & 2 + e^{x_1+x_2} \\ 2 + e^{x_1+x_2} & 2 + e^{x_1+x_2} \end{pmatrix} = (2 + e^{x_1+x_2}) \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ is semidefinite, thus $f_1(\mathbf{x})$ is convex. Since $x_1 - 3x_2 = 0$ is affine, we know that the problem is a convex optimization problem.

□

(b).

$$\begin{aligned}
& \min_{x_1, x_2} && x_1^2 + x_2^4 \\
& \text{s.t.} && x_1 e^{-(x_1 + x_2)} \leq 0 \\
& && x_1^2 - 2x_1x_2 + x_2^2 + x_1 + x_2 \leq 0 \\
& && 6x_1^2 - 7x_2 = 0
\end{aligned}$$

Proof. Since $6x_1^2 - 7x_2 = 0$ is not affine, we know that the problem is not a convex optimization problem. □